Current lead optimisation

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1: Overview

A power supply is needed for the superconducting magnets - this has to go from room temperature at the warm end, to a cooled temperature inside the cryostat. The heat leak from the current leads largely determines the running cost of the magnet system. Therefore, we want to minimise heat conduction and electrical resistance along the lead, by determining the optimum shape given the operating current and material. We can also derive the heat profile along the lead to see how this varies in a steady state.

Much of the derivation and results validation is based on the material in *Superconducting Magnets* "(Wilson, 1983, §11.1)".

2: Intuitive picture

We consider the situation where a steady state has been reached, and the temperature at each point of the current lead is stable. Therefore, we don't need the heat equation because we're not considering the temperature evolution at each point of the lead. Heat is conducted along the wire from the warm end to the cold end, and ohmic heating is produced due to resistance in the lead.

The coolant boiling off from the cryostat may be used to cool the lead - this gas moves from the cold to the warm end of the lead. In this case, we assume that because we're in a steady state, the gas and lead temperatures match at all points along the lead, so that as the gas passes the lead it absorbs heat and warms up.

Alternatively, we can consider a version that doesn't use gas cooling (i.e. cryogen-free system) - in this case there is only conductive cooling along the cable.

The optimum solution (in terms of minimising heat leak) is expressed in terms of IX/A, which represents the product of current and lead length, divided by cross-sectional area. This shows the trade-off involved in terms of the geometry across and along the cable, and the amount of current supplied.

For much of the derivation of the optimum, a rebased variable (z) is used to simplify the algebra. This represents the current multiplied by accumulated thermal resistance (i.e. temperature change given power) along the lead.

3: Theoretical background

This section outlines the steps taken to derive the optimum solutions, which follows *Superconducting Magnets* "(Wilson, 1983, §11.1)".

For both problems (convection and conduction cooled, conduction cooled), the optimum is first derived with respect to a rebased variable (z) to simplify the algebra. This represents the current multiplied by accumulated thermal resistance (i.e. temperature change given power) along the lead, rather than just the length. The length can then be subsequently obtained by integrating the thermal conductivity of the chosen material using the rebased variable.

In the gas cooled case, the optimum value is solved numerically, whereas in the conduction cooled case the solution has a closed form. The temperature profile along the lead can be calculated using intermediate equations. The accumulated thermal resistance can then be adjusted away from the optimum, to see how this affects the amount of current in the lead and maximum temperature along the lead.

3.1: Conduction and convection cooled derivation

3.1.1: General setup

Unless otherwise stated, all references in this section relate to *Superconducting Magnets* " (Wilson, 1983, §11.1)".

We start by assuming the system is at thermal equilibrium, so the various heat sinks / sources can be set to zero:

- Energy in (from the warm end) minus energy out (to the cold end and into the gas) plus energy generation inside = 0
- Energy in = $kAd\theta/dx$ (Fourier's law - small area or low thermal conductivity implies large temperature gradient)
- Energy out = $-kAd\theta/dx + H$ (But $d\theta/dx$ different at this point versus for energy in if gradient changes along the lead)
    - H is the gas cooling term. We can use equilibrium to convert H using the specific heat equation (i.e. $\dot{m}C_p d\theta$ with f representing efficiency of heat transfer, and noting we consider kg/s gas boil off to use this in terms of W)
- Energy generation = $\frac{I^2\rho}{A}dx$ (using $R = \rho\frac{L}{A}$, to use property of material itself)
- k and ρ are dependent on temperature, and we can link them using the Wiedemann-Franz law: $k(\theta)\rho(\theta) = L_0\theta$
    - The Lorenz number, l_0 reduces at lower temperatures for purer metals, but we have assumed it's fixed here, consistent with Wilson
- The whole equation can be divided through by dx to simplify (and to setup how this changes along the lead) "(Wilson, 1983, p. 257, eq. 11.3)":

$$\frac{d}{dx}(k(\theta)A\frac{d\theta}{dx}) - f\dot{m}C_P\frac{d\theta}{dx} + \frac{I^2\rho(\theta)}{A} = 0 \tag{1}$$

A few variables need to be substituted to tidy up the equation:

- We rebase distance x to current times accumulated thermal resistance, using $dz = \frac{Idx}{k(\theta)A}$ (remember this using result we're looking for of IX/A as well)
- We can use this to find that:

$$\frac{d\theta}{dz} = \frac{d\theta}{dx}\frac{dx}{dz} = \frac{d\theta}{dx}\frac{k(\theta)A}{I} \tag{2}$$

- Replace $k(\theta)\rho(\theta)$ with $L_0\theta$ (Wiedemann-Franz law)
- And set $w = \dot{m}C_L$, $u = \frac{C_P}{C_L}$, where C_L is the latent heat of evaporation of liquid helium

This results in the following 2nd order ODE "(Wilson, 1983, p. 258, eq. 11.8)":

$$\frac{d^2\theta}{dz^2} - \frac{fwu}{I}\frac{d\theta}{dz} + L_0\theta = 0 \tag{3}$$

We then consider the solution with complex conjugate roots (i.e. a damped oscillator with $b^2 - 4ac < 0$, or $\frac{fwu}{I}^2 - 4L_0 > 0$), and plug in the boundary conditions (i.e. $\theta = \theta_2$ at $z = Z_2$, and simplifying to $\theta = 0$ at $z = 0$), from which we get the following "(Wilson, 1983, p. 259, eq. 11.12)":

$$\theta = \theta_2\frac{e^{\alpha z}sin(\beta z)}{e^{\alpha Z_2}sin(\beta Z_2)} \tag{4}$$

Where:

$$\alpha = \frac{fwu}{2I}, \beta = \sqrt{L_0 - \alpha^2} \tag{5}$$

We set heat in equal to heat out at the cold end of the cable (conducted heat versus heat leak, $kAd\theta/dx = w$) using Fourier's law again, divide this by current (I) and set this equal to the derivative of Equation 4 using the result in Equation 2. Setting $z = 0$ after differentiating Equation 4 we obtain "(Wilson, 1983, p. 259, eq. 11.13)":

$$\frac{kA}{I}\frac{d\theta}{dx} = \frac{d\theta}{dz} = \frac{\beta\theta_2}{e^{\alpha Z_2}sin(\beta Z_2)} = \frac{w}{I} \tag{6}$$

3.1.2: Optimal solution

We then differentiate Equation 6 with respect to Z_2 (to minimise w/I), and rearrange to get $\alpha = -\beta\left(tan(\beta Z_2)\right)^{-1}$. By equating w/I in Equation 5 and Equation 6 we can find an alternative expression for α. These expressions can be equated and α is repeatedly substituted for the alternative βZ_2 expression to obtain "(Wilson, 1983, p. 259, eq. 11.15)":

$$\beta Z_2 = tan(\beta Z_2)\, ln(\frac{-2cos(\beta Z_2)}{fu\theta_2}) \tag{7}$$

- The cos appears when take $\frac{sin}{tan^{-1}}$
- A value for βZ_2 can then be found numerically once values for f, u and θ_2 have been set

The next equation is obtained by noting the following "(Wilson, 1983, p. 259, eq. 11.16)":

- $cos(x) = \frac{+}{-}(1 + tan^2(x))^{-1/2}$
- $tan(\beta Z_2) = -\beta/\alpha$ from above, and this value must be negative
- $\beta = \sqrt{L_0 - \alpha^2}$, from Equation 5
- $sin(\beta Z_2)$ must be positive to use practical values of θ, from Equation 4
- $cos(\beta Z_2)$ must be negative

$$cos(\beta Z_2) = -(1 + tan^2(\beta Z_2))^{-\frac{1}{2}} = -\alpha / L_0^{\frac{1}{2}} \tag{8}$$

We can then obtain the optimum heat leak by combining Equation 5 and Equation 8:

$$w = -\frac{2IL_0^{1/2}}{fu}\ cos(\beta Z) \tag{9}$$

At this point Z_2 can be obtained from the solved values of βZ_2 (using Equation 7) and β (by first obtaining α from Equation 8 and then β from Equation 5). The temperature along the lead can then be calculated by varying z in Equation 4.

To calculate temperature versus physical distance, we need the temperature-dependent thermal conductivity of the lead and then can rearrange Equation 2 to get $\frac{Ix}{A} = \int k(\theta)\, dz$. This gives a total length, including current and area: IX_2/A

3.1.3: Non-optimal solutions

Here we consider the case where Z_2 is increased from it's optimum (i.e. we take $Z_2/Z_{2_o} > 1$). For this we can't rely on Equation 7 onwards to determine βZ_2, but we can adjust the value of βZ_2 until w/I in Equation 5 and Equation 6 above are equal:

$$\frac{2\alpha}{fu} = \frac{\beta\theta_2}{e^{\alpha Z_2} sin(\beta Z_2)} \tag{10}$$

- For a given value of βZ_2, we can find β and then α follows from Equation 5
- f, u and θ_2 are inputs

This value of βZ_2 is used in Equation 4 to calculate the temperature along the lead.

We can again integrate $\int k(\theta)\, dz$ to get a revised value of IX_2/A. Assuming length and cross-sectional area are fixed, we can then find how this increase from optimum Z_2 impacts the

current in the lead: $\frac{IX_2/A}{I_oX_2/A} = \frac{I}{I_o}$.

3.2: Conduction cooled version

3.2.1: General setup

This follows the same steps but with $H = 0$, and so $\alpha = 0$ and $\beta = \sqrt{L_0}$ (this is the same as setting $f = 0$)
Supplying the boundary conditions (and using the actual temperature at the cool end) we can get values for D and E in terms of θ_1 and θ_2 "(Wilson, 1983, p. 259, eq. 11.9)":

- $E = \theta_1$
- $D = \frac{\theta_2 - \theta_1 cos(\beta Z_2)}{sin(\beta Z_2)}$

We then get the following equation for temperature (this is the equivalent of Equation 4):

$$\theta = \frac{\theta_2 - \theta_1 cos(\beta Z_2)}{sin(\beta Z_2)} sin(\beta z) + \theta_1 cos(\beta z) \tag{11}$$

We then differentiate with respect to z to equate this to w/I (this is the equivalent of Equation 6):

$$\frac{kA}{I}\frac{d\theta}{dx} = \frac{d\theta}{dz} = \beta \frac{\theta_2 - \theta_1 cos(\beta Z_2)}{sin(\beta Z_2)} = \frac{w}{I} \tag{12}$$

3.2.2: Optimal solution

We then differentiate Equation 12 with respect to Z_2 (to minimise w/I) and rearrange to find a closed form solution for Z_{2_o}:

$$Z_{2o} = \frac{1}{L_0^{1/2}} arccos(\frac{\theta_1}{\theta_2}) \tag{13}$$

- We use the identity $sin^2(x) + cos^2(x) = 1$
- And $\beta = \sqrt{L_0}$
- This also gives us the result that $cos(\beta Z_2) = \frac{\theta_1}{\theta_2}$ and $sin(\beta Z_2) = \sqrt{1 - \frac{\theta_1^2}{\theta_2^2}}$

  We can then calculate the optimal heat leak by substituting the value of βZ_2 back into Equation 12:

$$\frac{w}{I} = \sqrt{L_0(\theta_2^2 - \theta_1^2)} \tag{14}$$

  We can use a simplified form of Equation 11 to calculate the temperature along the lead:

$$\theta = \frac{\theta_2 - \theta_1 cos(\beta Z_2)}{sin(\beta Z_2)} sin(\beta z) + \theta_1 cos(\beta z) = \sqrt{\theta_2^2 - \theta_1^2}\, sin(\beta z) + \theta_1 cos(\beta z) \qquad (15)$$

Again, to translate Z_2 into a physical length, we need to use temperature-dependent thermal conductivity data for the lead material, and determine how the cross-sectional area changes.

3.2.3: Non-optimal solutions

We increase Z_2 above optimum, and $\beta = \sqrt{L_0}$ is unchanged. The heat leak in this case can be calculated from Equation 12.

4: Interactive models

A spreadsheet has been produced to model results for the following:

- Convection and conductive cooling, conduction cooling only
- Annealed pure copper and phosphorus-deoxidised copper (referred to as Cu and P-Cu below), using thermal conductivity charts in "Superconducting Magnets" combined with Wiedemann-Franz "(Wilson, 1983, p. 262, Fig. 11.4)"
  The model assumes the following:
- Steady state with thermal equilibrium (in practice, central solenoid / PF coils would be pulsed)
- Perfect heat exchange assumed for the gas ($f = 1$)
- Lorenz number L_0 fixed in Wiedemann-Franz (this gives temperature-agnostic result)
- Fixed cross-sectional area of the lead (doesn't affect the optimum but does affect conversion back to physical distance)
- Using $\theta_1 = 0$ rather than actual cold end temperature (4K vs 0K very little difference)
- Constant specific heat for helium
- 1 dimensional
- No magnetoresistance - would be a large effect on copper with magnet nearby

5: Results validation

The following high-level observations have been made:

- In the optimum case, the temperature gradient tends to zero as it reaches the warm end
- For the same cold / warm end temperatures, the conduction cooled case results in a much lower optimum IX/A - there is less cooling to the surroundings, so it's less able to transport away ohmic heating

The optimum heat leak / current has been calculated as 1W/kA and 47W/kA in the gas and conduction cooled cases respectively - i.e. for the same current, the heat leak is much higher in

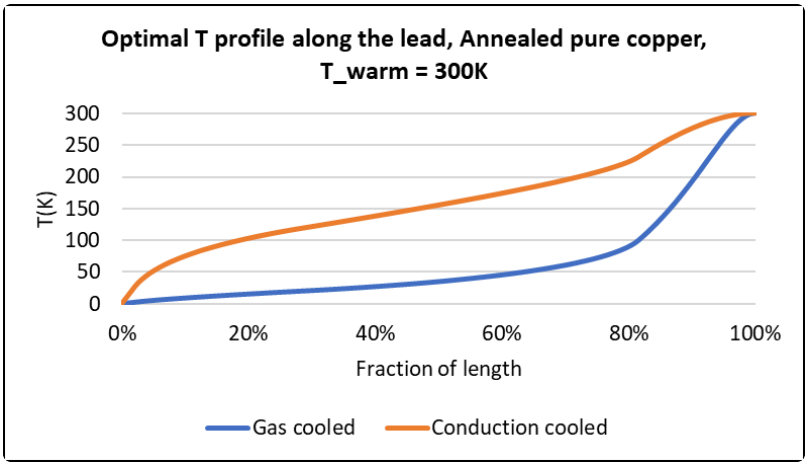
the absence of convective cooling. This matches figures published by Wilson "(Wilson, 1983, p. 260)" and Ballarino "(Ballarino, 2014)".

The table below compares optimum $IX/A \times 10^{-6}$ as the cooling and material types are varied:

	Cu	P-Cu	Ratio
Gas cooled	24.2	3.3	725%
Conduction cooled	5.4	1.7	313%
Ratio	22%	52%	

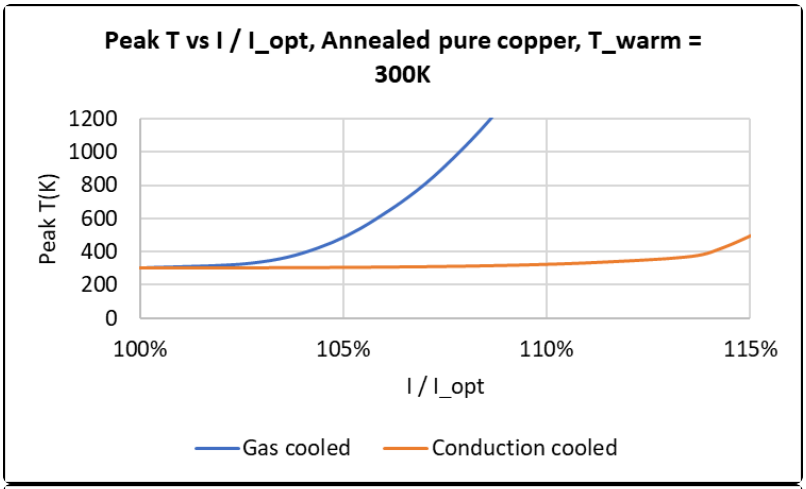
- If we assume the cross-sectional area is unchanged, this implies purer copper can use a higher current and/or longer cable to achieve the same heat leak relative to current (though if current is higher, the absolute heat leak will be greater than for less pure copper).
- This is due to the higher thermal conductivity of the purer copper (and lower resistivity).
- The gas cooled figures broadly tally with those from Wilson "(Wilson, 1983, p. 263)"

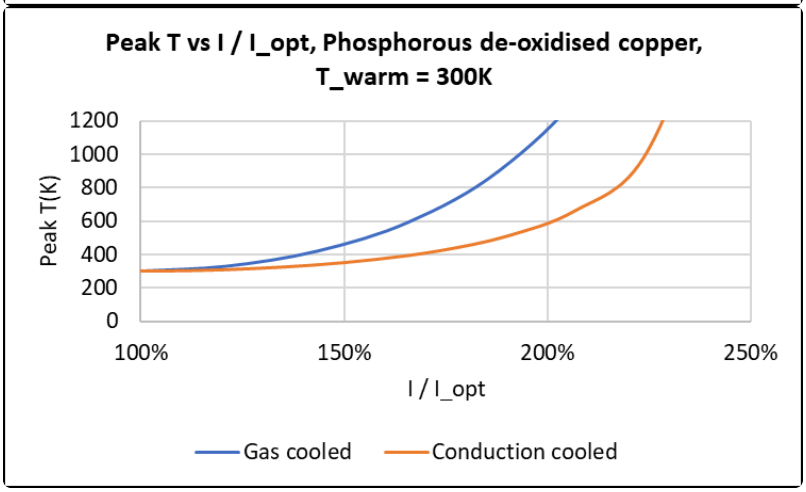
The chart below compares the temperature profile along the leads when using Cu and the temperature at the warm end is 300K:



- This visibly shows that the gas cooling enables the temperature to be kept much lower, much further along the cable.

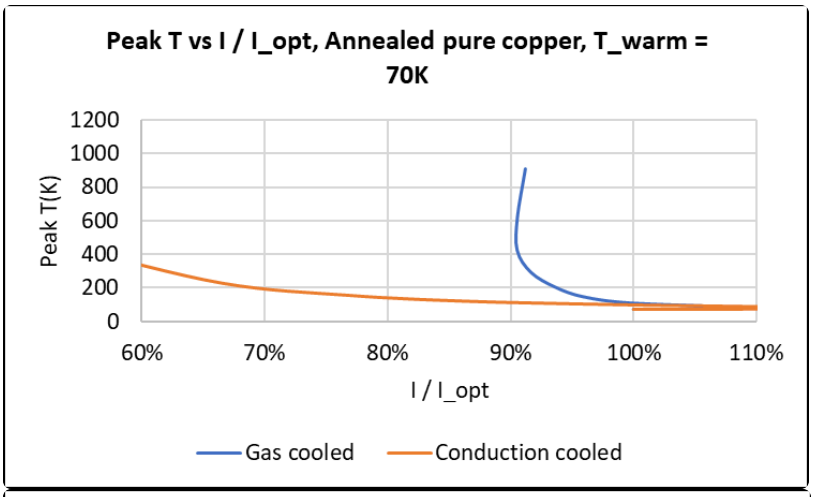
The charts below show how the peak temperature changes as the current is increased above optimal (I/I_{opt})

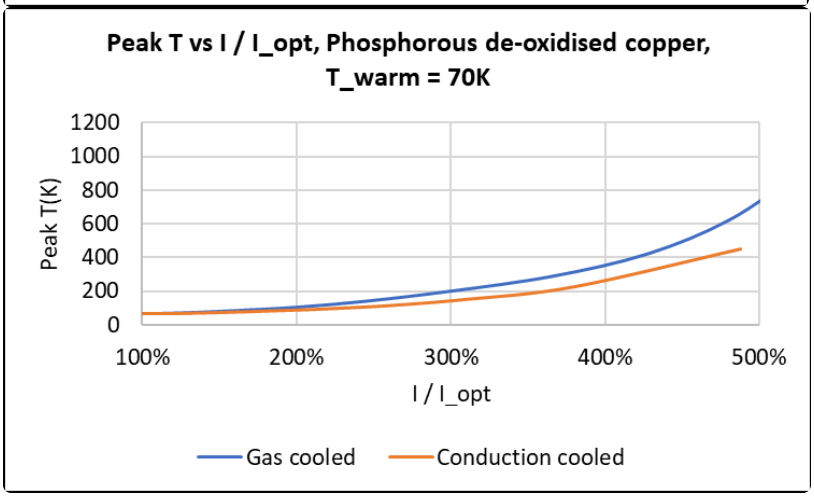




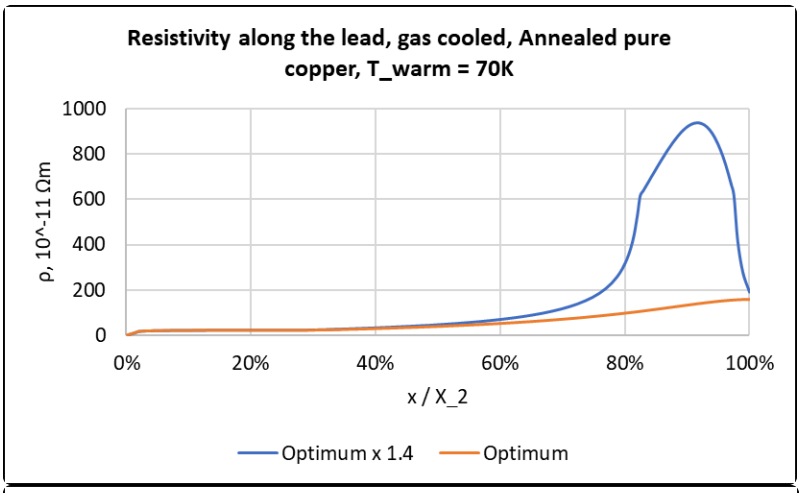
- Note the much larger range of currents that are modelled for P-Cu (different x-axis scales have been used); as observed by Wilson, the peak temperature in Cu cables is very sensitive to changes in current.
- The gas cooled lead is also more sensitive to changes in current than the purely conduction cooled version.

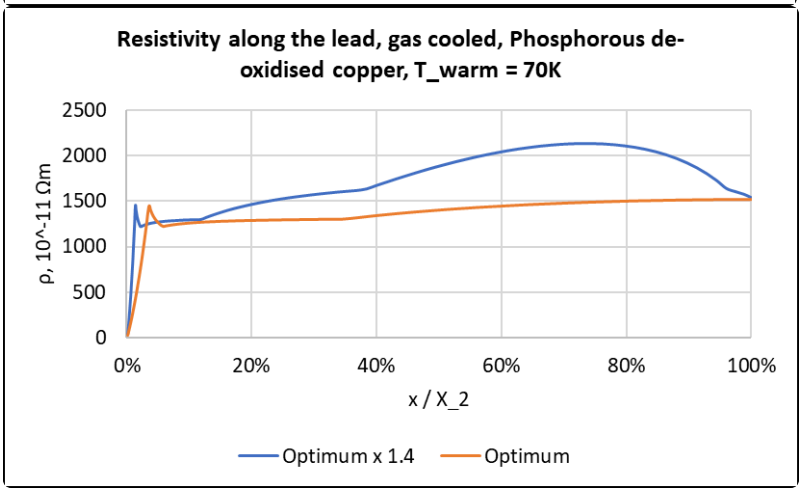
For a Cu lead, if the temperature at the warm end is reduced (to e.g. 70K), the current actually reduces when accumulated thermal resistance is increased above 140% or so of the optimum:





This doesn't occur for P-Cu, where the peak temperature rises monotonically with current, as in the 300K charts. I haven't yet confirmed the cause, but the effect sits in the 20-70K range, where the purer copper's properties change most rapidly with temperature and where the assumption of a fixed Lorenz number is least accurate. The results may be due to this assumption.





6: Open questions

The following are the areas I would explore further, given more time:

- Confirming whether the current reduction for pure Cu with a 70K warm end is due to a fixed Lorenz number, by re-running with measured resistivity data rather than resistivity derived via Wiedemann-Franz
- Checking the impact of reducing the Lorenz number for lower temperatures (5-100K), particularly for the purer copper - resistivity has been derived by assuming the Lorenz number is equal to the Sommerfeld limit, but in reality it will reduce significantly at low temperatures
- Extending the model to consider a superconducting power supply that is split into 2 sections (the colder section using HTS, and the warmer section using a conventional lead)
    - Can then tweak the temperature where the sections meet to see if this can also be optimised
    - Will need to ensure the current is below the HTS critical current

7: Key takeaways

These results suggest the following:

- A higher current or longer cable can be used when made using purer copper
- But, the temperature of these cables is more sensitive to excess current - so a larger operating margin might be required
- This also applies when using gas cooling (rather than just conductive cooling); a larger current or longer cable can be used, but this is again more sensitive to fluctuations
- A gas cooled lead has a heat leak about 50x smaller (versus current) than conduction cooled only
- This should be optimised for the actual operation cycle of the magnet - this only considers a single current (HTS leads or intermediate heat sinks could be used to reduce the heat leak when operating at zero / low current)
- There's a trade-off between length and area of the cable

8: References

[1] M. N. Wilson, *Superconducting Magnets*, Clarendon Press, Oxford, 1983
[2] A. Ballarino, “Current leads, links and buses”, CAS–CERN Accelerator School: Superconductivity for Accelerators, CERN-2014-005, pp. 547–558 (arXiv:1501.07166)